Panskura Banamali College (Autonomous)

B.Sc.(H) 4th Semester Examination -2021 (Online Mode)

Subject: Computer Science **Paper: CC10P**

Paper Name: Database Management Systems Lab

F.M.: 20 **Time : 2 Hours**

Instruction

[Make your answer script into PDF and MP3 file format and file name as like candidate_id-paper_code.pdf . For Example – candidate_id: 2020UG009967, paper_code : CC10P , then file name will be :: 2020UG009967- CC10P.pdf for question 'A' and 2020UG009967- CC10P.mp3 for question 'B']

A. Answer any one question: **1 x12 = 12**

1. BOOK(acc_no, title, publisher, no_of_copies, subject_category)
BORROWER(card_no, b_name, b_address)
ISSUE(acc_no, card_no,date_of_issue, date_of_return)
SUPPLIER(s_name, acc_no, price)

a) *List the names of the borrower who have issued some books with title containing the word 'database'.*

b) *List the publisher of all books in the library such that the name of publisher appears only once.*

c) *List the accession number of all books issued between March 2014 and August 2014.*

d) *List the accession number and titles of all books containing less than five copies.*

e) Display book and issue details.

2. Consider the following database :
hotel (h_no, h_name, h_address)
room (r_no, h_no, type, charge)
booking (h_no, g no, date_from, date_to, r_no)
guest (g_no, g_name, g_address)
(a) List all single rooms with a charge below rupees 2000.
(b) List all unoccupled room for the 'Grand hotel'.
(c) List the guest details currently staying in 'Seahawk' hotel.
(d) List the details of all guests staying at the 'Grand hotel from 26th June 2013

3. Consider the following database :
Project (Project#, P_name, Chief-Architect)
Employee (Emp#. E_name)
(a) Get details of all employees in descending order.
(b) Get Emp# of employees working on project no 353.
(c) Get details of employees working on more than two projects.
(d) Get details of employees working on a project named 'Database'.

4. Consider the following database : Employee (e no, e name, e addr)
Project (p no, p name)
Work(e no,p no)
(a) Find the number of employees who are working on more than one project.
(b) Print name of the employees who are working on project name 'banking system'.
(c) Print the name of the employees who are not working in any project.
(d) Display all information about the employees whose 2nd character of name is either i', 'or'. 'o'.

5. Consider the following database:
Emp(e_no,e_name,basic_sal,D_O_J)
Dept(e_no,d_no,d_name)
(a) Display the name of the employee who gets 2nd highest salary.
(b) Find department_wise total salary.
(c) Find the no. of employees in 'Finance' department.
(d) Increase salary by 10% for every employee.

6. Consider the following database:
hotel(h_no,h_name,h_address)
room(r_no,h_no,type,charge)
booking(h_no,g_no,date_from,date_to,r_no)
guest(g_no,g_name,g_address)
(a) List all single rooms with a charge below rupees 2000.
(b) List all unoccupied room for the 'Grand hotel'.
(c) List the guest details currently staying in 'Seahawk' hotel.
(d) List the detals of all guests staying at the "Grand' hotel from 26th June 2013.

7. Client_Master(cl_no,cl_name,cl_city,due_balance)
Product_master(P_no,P_name,Profit_Percentage,self_price)
Sales_Man_Master(E_no,E_name,E_amount)
a) Select all the client having 1st letter ‘S’ in his name.
b) Change the city ‘chennai’ in space cl_no=’cl05’
c) Find the product name whose profit_percentage is maximum.
d) Display the E_name whose E_no is odd and E_amount is greater than 1500 order by E_amount in descending.
e) Display profit details of the product “cocacola”

8. SUPPLIER(sno,sname,status,city)
PARTS(pno,pname,color,weight,city)
PROJECT(jno,jname,city)
SPJ(sno,pno,jno,qty)
a)select sname values for suppliers who supply project j1;
b) get sno values for suppliers who supply project j2 with parts p3
c) get jname values for project supplied by supplier s1.
d) get color values for parts supplied by supplier APU.
e) get jnmes for those project which are supplied by supplier RAM.

B. Answer any four questions: **4x2=8**

[ONLINE Viva-Voce : Record the audio of your answers as like .mp3 format]

1. What are different types of joins in SQL?
2. What is BCNF ?
3. What are the different types of languages that are available in the DBMS?
4. What is the purpose of SQL?
5. What are some examples of open source and commercial Relational DBMSs?
6. What is a database model? and name a few common database models?
7. What are the different levels of data abstraction?
8. What is normalization and de-normalization?
9. What are the main differences between Primary key and Unique Key?
10. Explain Entity, Entity Type, and Entity Set in DBMS?

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